SWERVE DRIVE SPECIALTIES

MK4c Swerve Module Assembly Guide



Table Of Contents

Overview **3**
- Swerve Module Configurations 3
- Assembly Guide Conventions 3
- Supplies needed 3
- Use of Threadlocker 4
- Lubrication 4

Step 1: Center Column Preparation **5**
Step 2: Steering Bearing Installation **6**
Step 3: Intermediate Shaft Installation **7**
Step 4: Center Column Assembly **8**
Step 5: Base Pulley Installation **9**
Step 6: Lower Drive Assembly **10**
Step 7: Upper Gear Installation **10**
Step 8: Encoder Installation **12**
Step 9: Motor Subassembly **13**
- Falcon 500 Motors 13
- Kraken X60 Motors 14
- NEO Motors 15
- NEO Vortex Motors (8mm Keyed Shaft) 16

Step 10: Motor Plate Installation **17**
Step 11: Wheel Subassembly **18**
- Billet Wheel 18
- Colson Wheel 19

Step 12: Wheel Installation **20**
Step 13: Swerve Module Installation **22**
Appendix **23**
- Frame Tube Dimensions 23

Overview

Swerve Module Configurations

The swerve module can be configured as follows:

- Motor Type: Falcon 500, Kraken X60, NEO, or NEO Vortex
- Gear Ratio: L1+, L2+, or L3+
- Wheel Type: Billet or Colson

Please note:

- *Motor Type* refers to the motor specific components provided in the kit, such as the pinion gears and shaft spacers. The motors are not included in the swerve module kit.
- The swerve module kit comes with the configurations selected when ordered. The kit does not include all of the components for all of the configurations.

Assembly Guide Conventions

This assembly guide details the assembly of all configurations, though the following configuration is primarily used for illustration purposes:

- Motor Type: Kraken X60
- Gear Ratio: L2+
- Wheel Type: Billet

Details of particular importance are indicated with **IMPORTANT**. Not following them could significantly impact the operation of the swerve module.

Supplies needed

Tools

- 1/8” hex key (for #10 button head screws)
- 5/32” hex key (for #10 socket head screws)
- 3/16” hex key (for 1/4” socket head screws)
- #1 phillips screwdriver (for the housing screws of a CTRE CANcoder or equivalent)

Overview, cont’d

Threadlocker and Retaining Compound

- For screws: Henkel Loctite 243 Threadlocker or equivalent
- For encoder magnet: Henkel Loctite 609 Retaining Compound or equivalent

Use of Threadlocker

IMPORTANT Use Henkel Loctite 243 Threadlocker or equivalent to secure all screws, except the screws used to secure the steering encoder, as it may adversely react with the encoder housing.

Lubrication

IMPORTANT Lubricate all gears with Lucas White Lithium, Lucas Red “N” Tacky, or equivalent grease BEFORE using the swerve module, and REGULARLY with use.

Step 1: Center Column Preparation

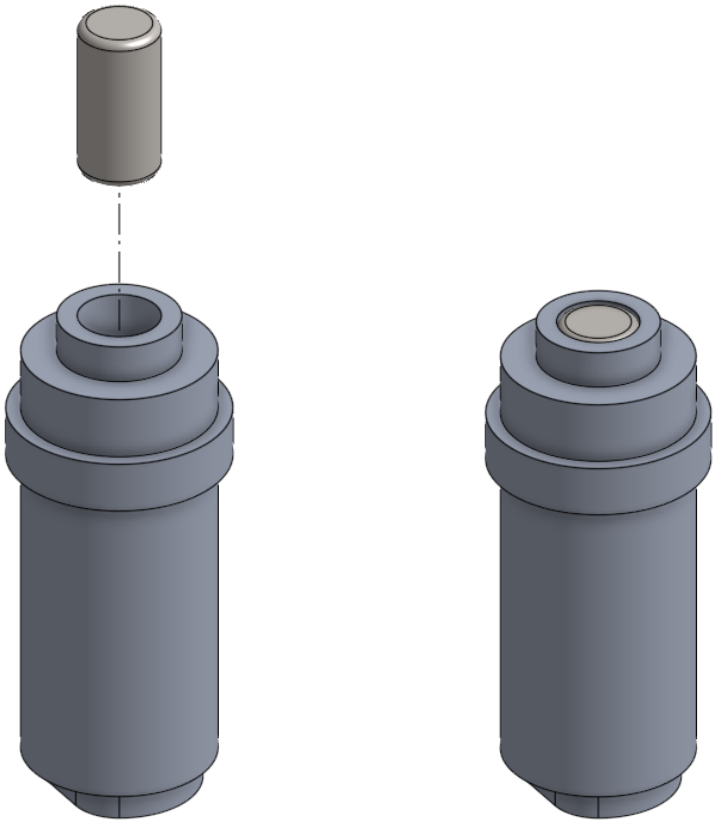
Bond the magnet to the Center Column:

1. Apply Loctite 609 or equivalent to the magnet.
2. Fully insert the magnet into the 0.25” hole in the Center Column.

Please note the magnet is included with the encoder, and not with the swerve module kit.

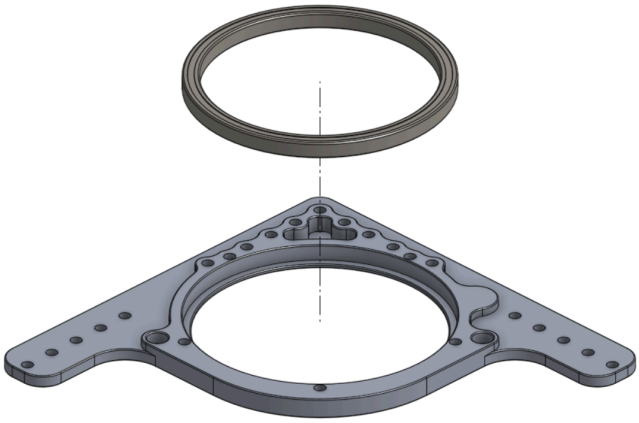
IMPORTANT

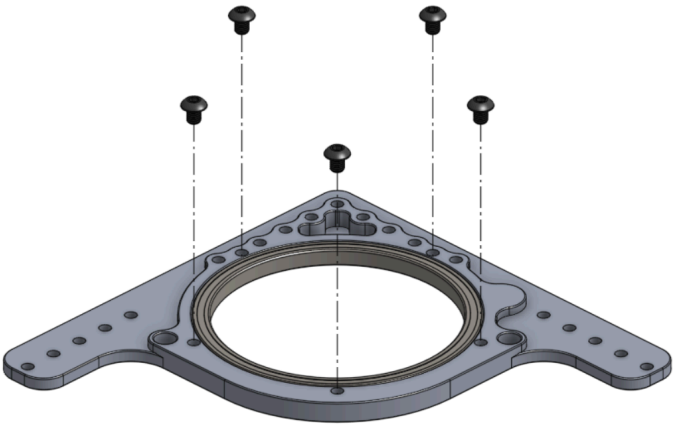
- Ensure the magnet is flush with the top of the Center Column.
- Wipe off any excess retaining compound from the magnet and Center Column.
- Do not use cyanoacrylate adhesive (“Super Glue”) to bond the magnet in place. It has a very short working time, and the magnet may bond in place before it gets fully inserted into the Center Column.
- Set the Center Column subassembly aside with the magnet facing upward so the retaining compound can cure without the magnet inadvertently shifting.



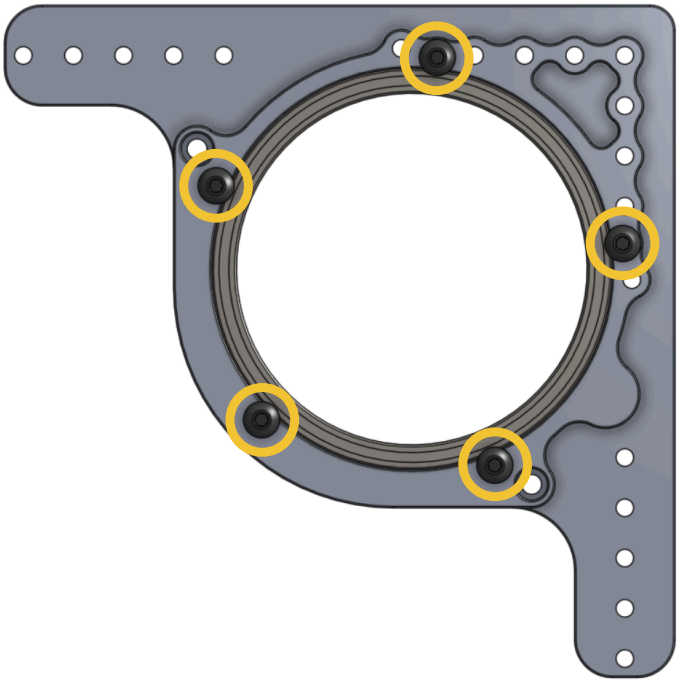
Step 2: Steering Bearing Installation

Install the large Steering Bearing into the Main Plate, and then secure it with five #10-32 x 0.25” Button Head Screws.





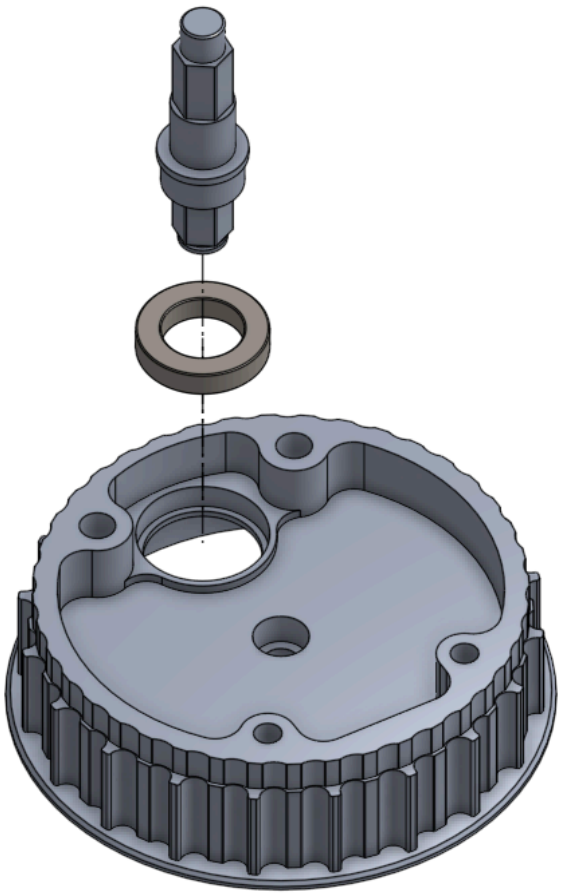
As the Main Plate has numerous threaded holes, ensure the correct five are used for securing the bearing.

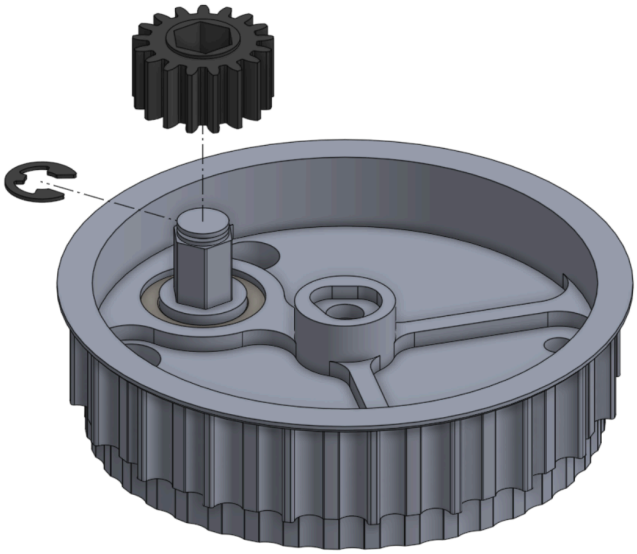


Step 3: Intermediate Shaft Installation

Insert the 6802ZZ Bearing (24mm OD) into the Base Pulley, and then the Intermediate Shaft into that Bearing.

Next, install the small 0.375” Hex Bore Gear onto the Intermediate Shaft, and then secure it by snapping the e-clip into the groove in the Intermediate Shaft. **IMPORTANT** Ensure the e-clip is fully seated in the groove.



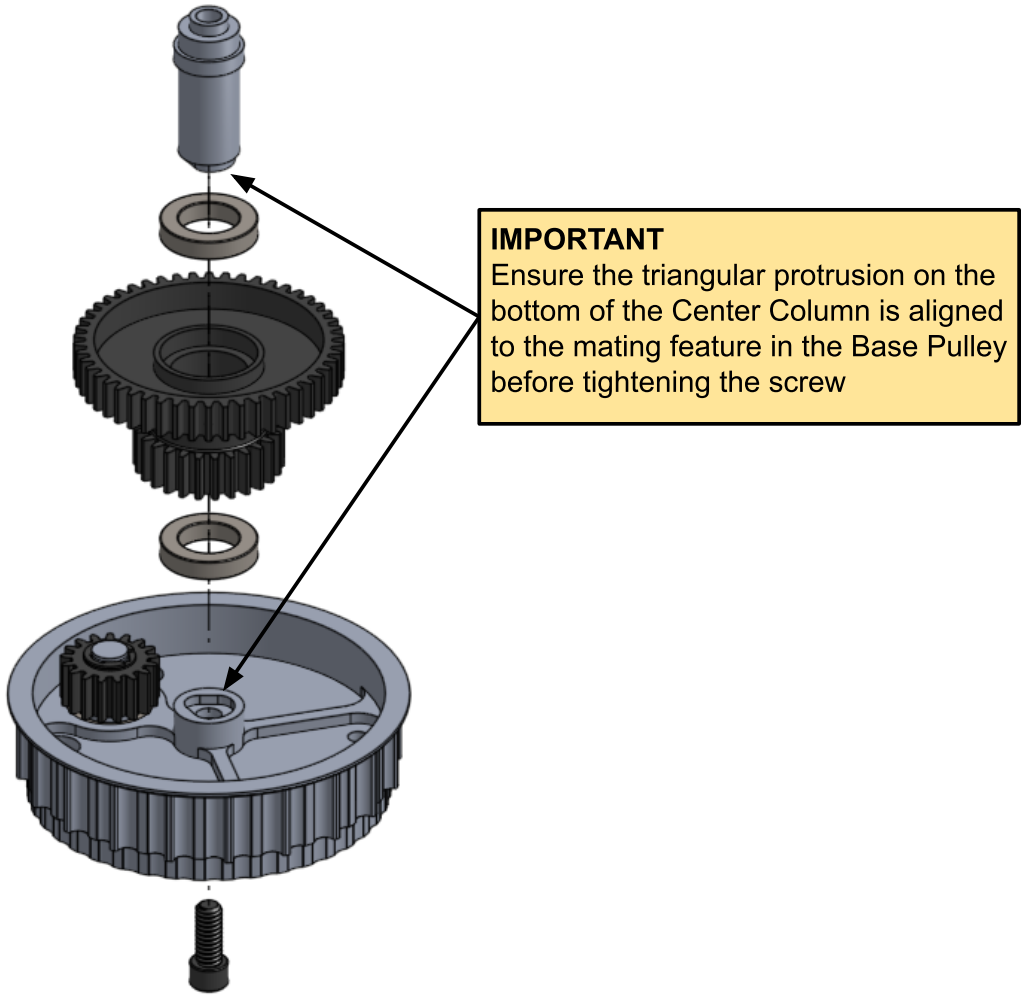


The size of that gear depends on the gear ratio the swerve module was configured with:

Gear Ratio Configuration	# Gear Teeth
L1+	19
L2+	17
L3+	16

Step 4: Center Column Assembly

Insert two 6802ZZ Bearings (24mm OD) into the Double Gear, and then capture those parts with the Center Column secured by a 1/4”-20 x 0.625” Socket Head Screw.

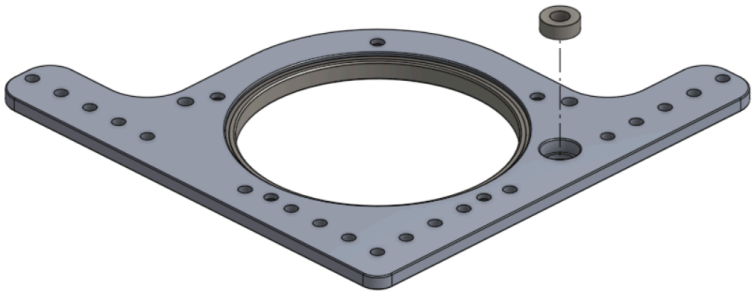


The size of the Double Gear depends on the gear ratio the swerve module was configured with:

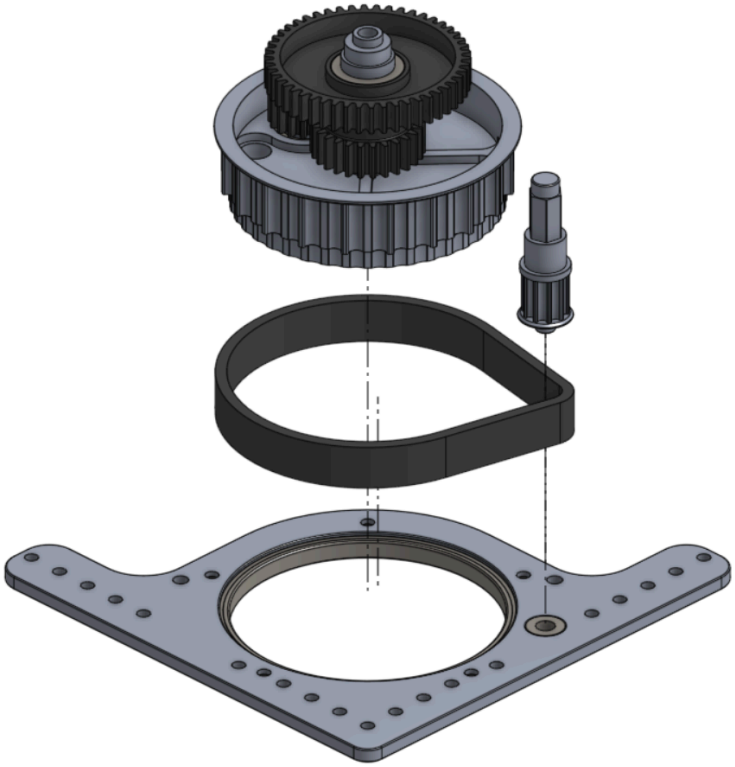
Gear Ratio Configuration	# Upper Gear Teeth	# Lower Gear Teeth
L1+	50	25
L2+	50	27
L3+	50	28

Step 5: Base Pulley Installation

Insert the R188ZZ Bearing (0.5” OD) into the Main Plate.



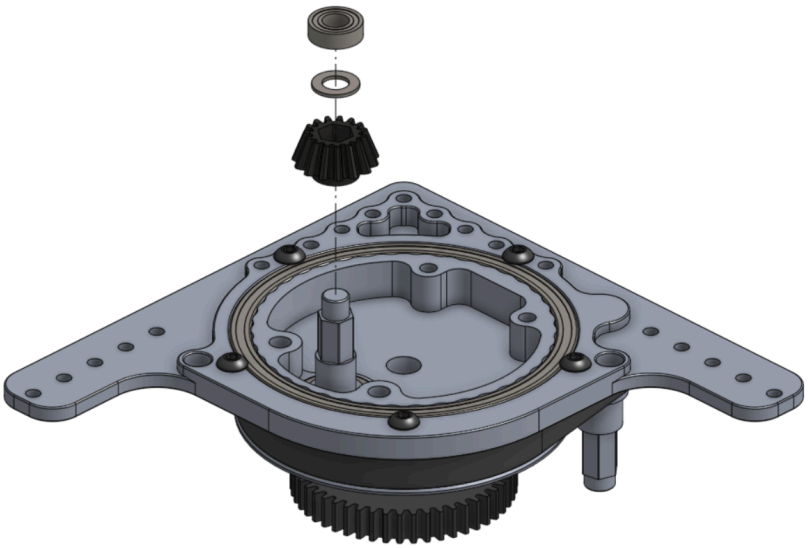
Place the Steering Belt around the Steering Shaft and Base Pulley, and then collectively insert the Base Pulley into the large bearing and the Steering Shaft into the small bearing.



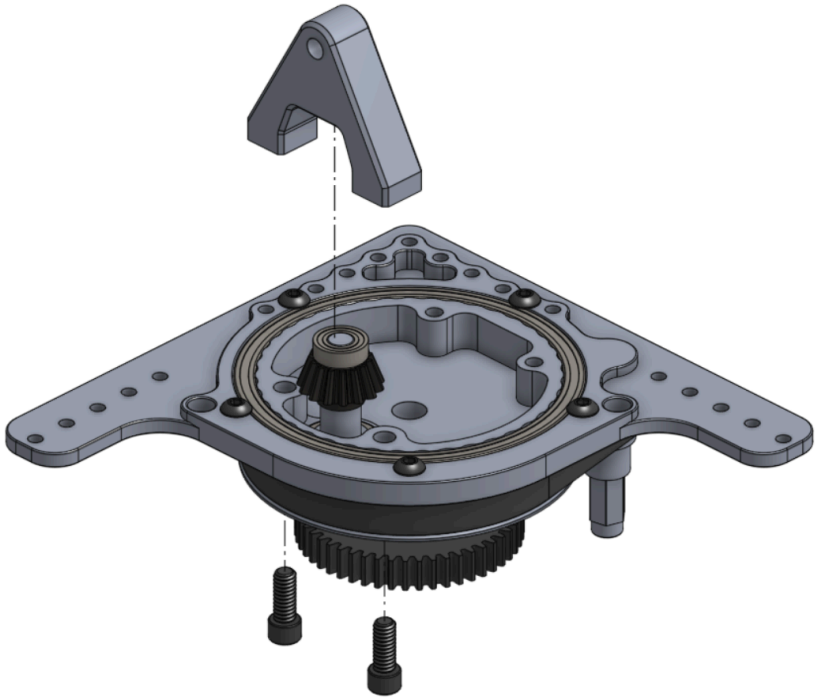
The bottom of the Base Pulley should be flush with the bottom of the large bearing.

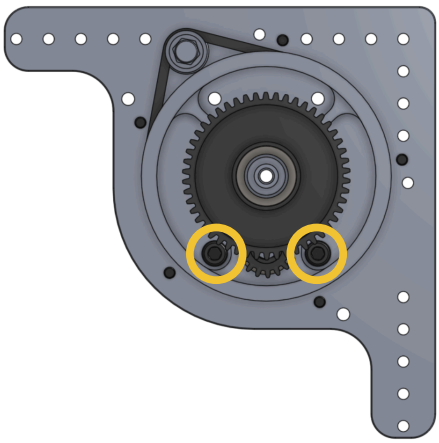
Step 6: Lower Drive Assembly

Place the 15T Bevel Gear onto the Intermediate Shaft, followed by the 1mm Thick Shim and then the 688ZZ Bearing (16mm OD).



Attach Wheel Mount B to the Base Pulley with two 1/4”-20 x 0.625” Socket Head Screws.

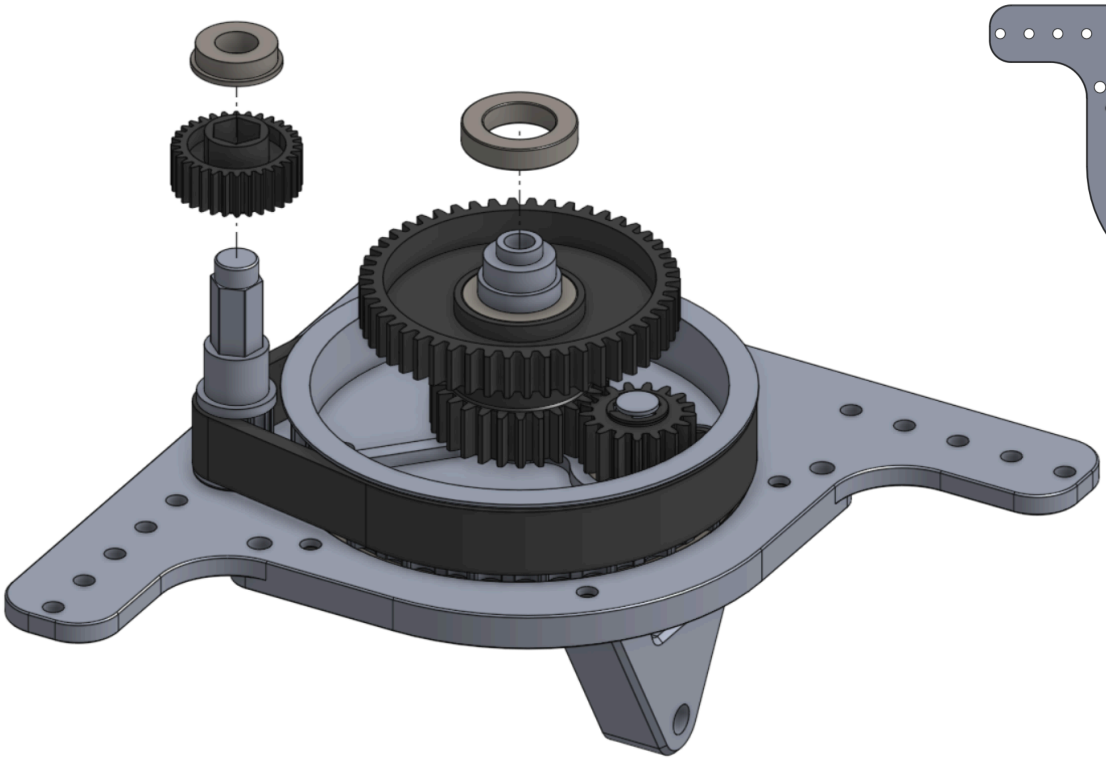




Step 7: Upper Gear Installation

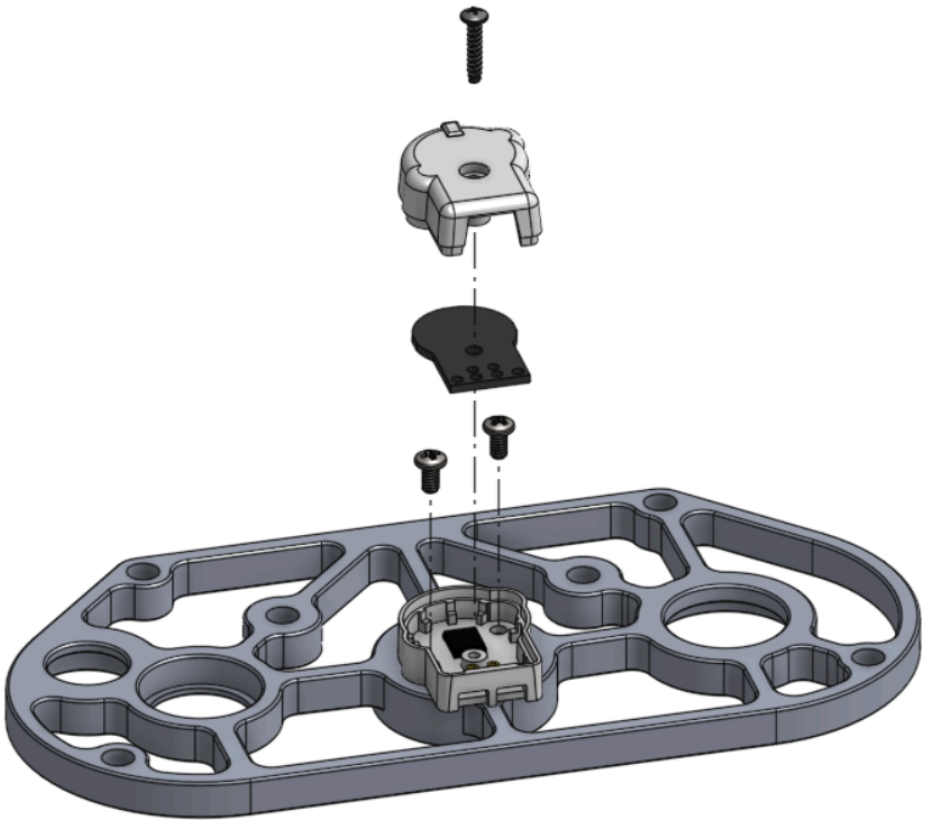
1. For the Steering Shaft: Place a 32T gear onto the Steering Shaft, followed by a F689ZZ Bearing (17mm OD). with the bearing's flange facing down.
2. Place a 6802ZZ Bearing (24mm OD) onto the Center Column.

.



Step 8: Encoder Installation

1. Secure the encoder housing's base to the Motor Plate with the two #3-48 screws included with the encoder. **IMPORTANT** Do not use threadlocker on those screws, as it may damage the housing.
2. Secure the circuit board and encoder housing's cover to the encoder housing's base with the #2-28 screw included with the encoder. **IMPORTANT** Do not use threadlocker on that screw, as it may damage the housing. **IMPORTANT** Do not overtighten that screw, as it may damage the housing.



Step 9: Motor Subassembly

The swerve module can be configured to work with a matching pair of Falcon 500, Kraken X60, NEO, or NEO Vortex motors. The swerve module can work with nonmatching motors (for instance a Kraken X60 drive motor and Falcon 500 steering motor), though additional parts for that arrangement may be needed.

The motor shafts need to be reduced in length by approximately 0.375” cut from Kraken X60 motor shafts and approximately 0.25” cut from Falcon 500, NEO, and NEO Vortex motor shafts so they do not interfere with other parts in the swerve module. Alternatively, SDS Motor Risers (available separately) can be used so the motor shafts can remain uncut. The following instructions are based on using the Motor Risers.

Falcon 500 Motors

Use the motor shaft spacers provided with the Falcon 500 motors.

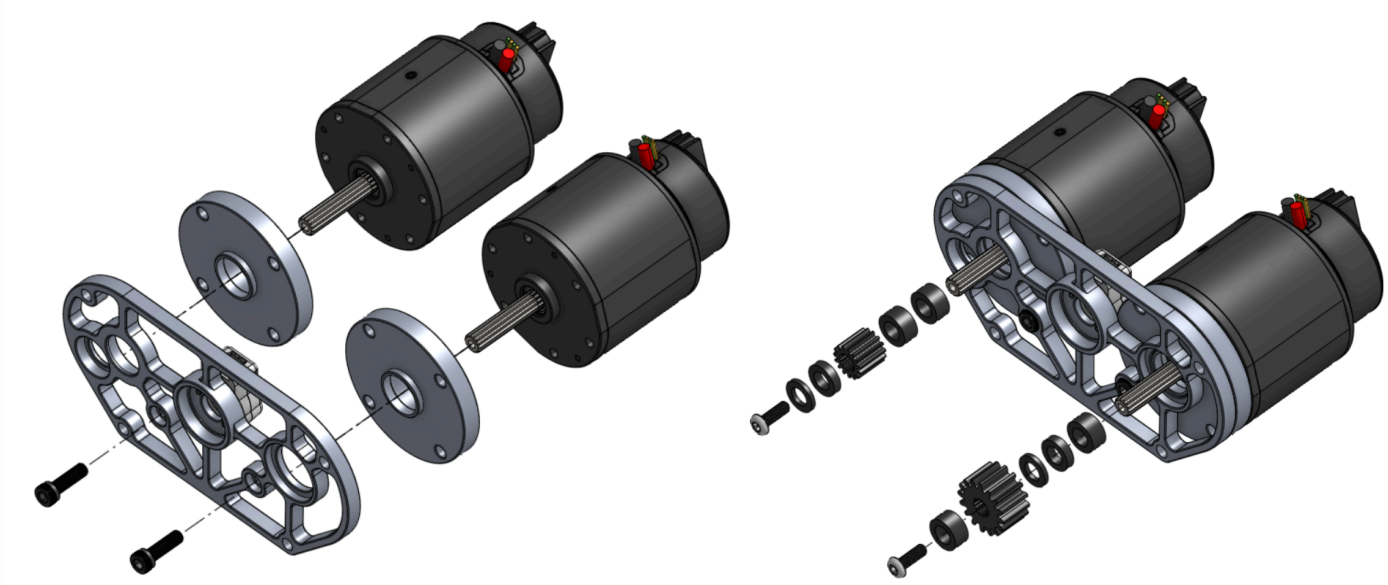
Secure the Motors and Motor Risers to the Main Plate with two #10-32 x 0.625” Socket Head Screws (included in the Motor Riser Kit). If no Motor Risers are used, then use the two #10-32 x 0.375” Socket Head Screws included in the swerve module kit.

Drive Motor

Install the following onto the motor shaft in order: 0.25” Long Spacer, 0.125” Long Spacer, 0.063” Long Spacer, 16T Pinion Gear, 0.25” Long Spacer. Secure those parts with the #8-32 x 0.5" Truss Head Screw. **IMPORTANT** Ensure threadlocker is used on that screw.

Steering Motor

Install the following onto the motor shaft in order: 0.25” Long Spacer, 0.25” Long Spacer, 15T Pinion Gear, 0.125” Long Spacer, 0.063” Long Spacer. Secure those parts with the #8-32 x 0.5" Truss Head Screw. **IMPORTANT** Ensure threadlocker is used on that screw.



Step 9: Motor Subassembly, cont'd

Kraken X60 Motors

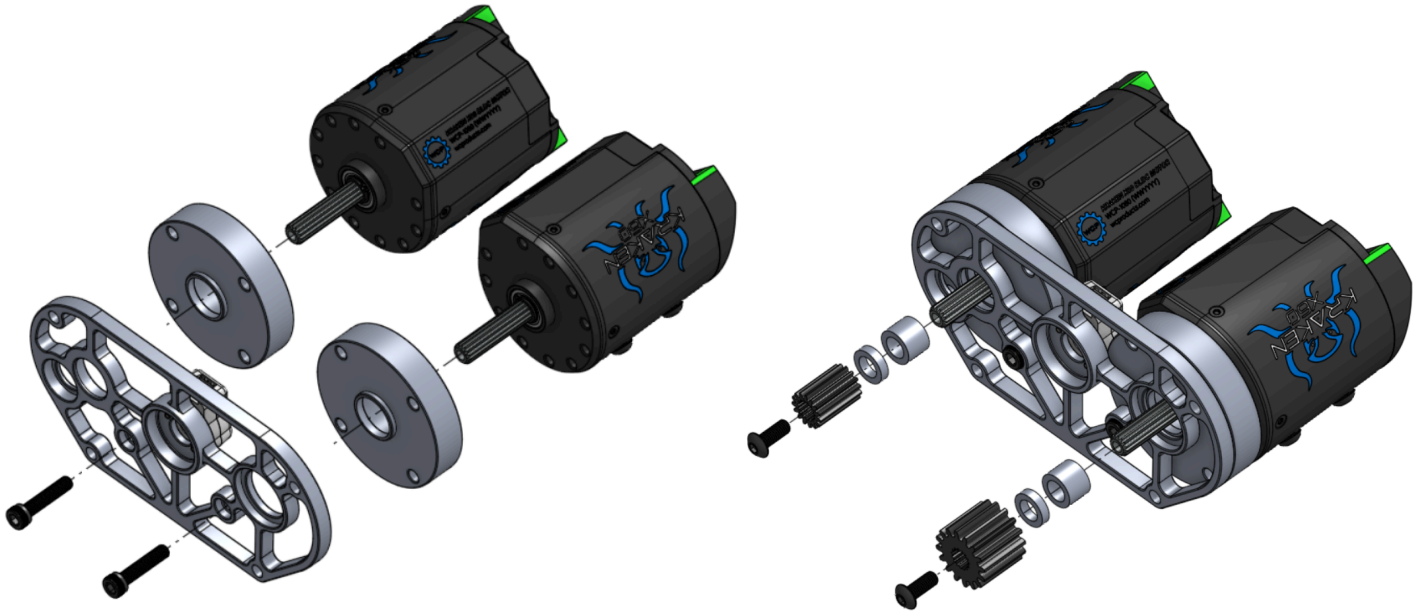
Secure the Motors and Motor Risers to the Main Plate with two #10-32 x 0.875" Socket Head Screws (included in the Motor Riser Kit). If no Motor Risers are used, then use the two #10-32 x 0.375" Socket Head Screws included in the swerve module kit.

Drive Motor

Install the following onto the motor shaft in order: 0.375" Long Spacer, 0.125" Long Spacer, 16T Pinion Gear. Secure those parts with the #10-32 x 0.5" Button Head Screw. **IMPORTANT** Ensure threadlocker is used on that screw.

Steering Motor

Install the following onto the motor shaft in order: 0.375" Long Spacer, 0.125" Long Spacer, 15T Pinion Gear. Secure those parts with the #10-32 x 0.5" Button Head Screw. **IMPORTANT** Ensure threadlocker is used on that screw.



Step 9: Motor Subassembly, cont’d

NEO Motors

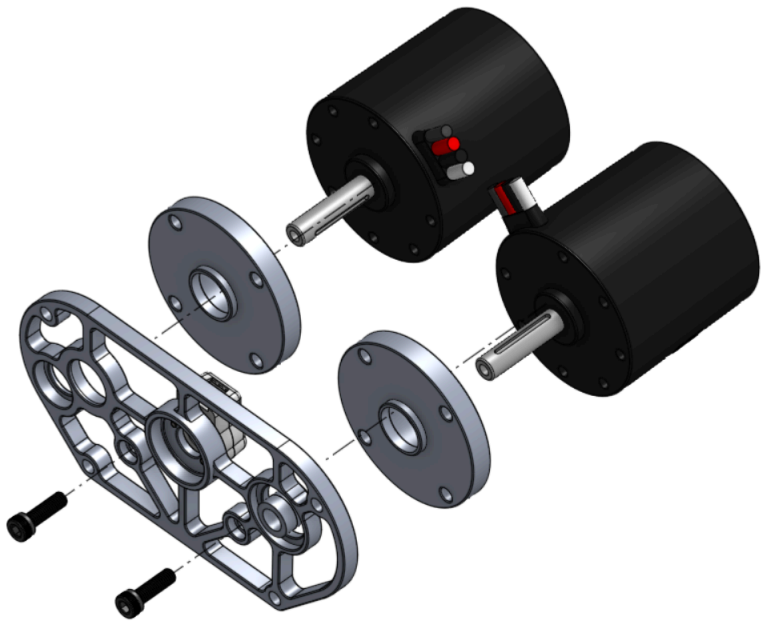
Secure the Motors and Motor Risers to the Main Plate with two #10-32 x 0.625” Socket Head Screws (included in the Motor Riser Kit). If no Motor Risers are used, then use the two #10-32 x 0.375” Socket Head Screws included in the swerve module kit.

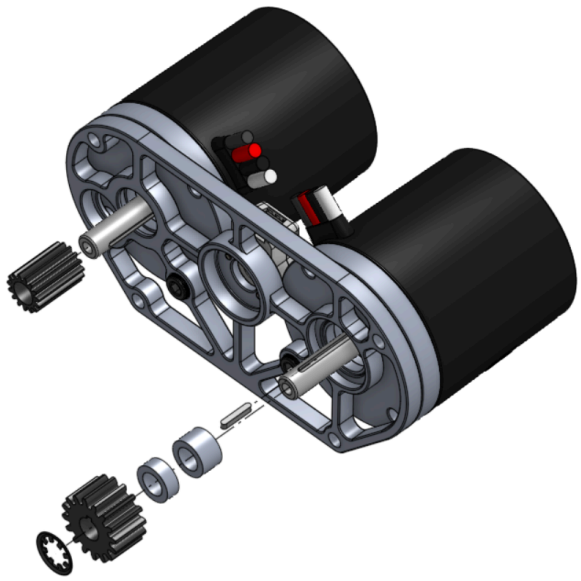
Drive Motor

Install the following onto the motor shaft in order: 0.313” Long Spacer, 0.188” Long Spacer, 2mm Machine Key, 16T Pinion Gear. Secure those parts with the 8mm ID Retaining Ring.

Steering Motor

Install the 15T gear (may be slip or press fit) 0.125” from the end of the shaft, using Loctite 609 Retaining Compound to fully secure the gear in place.





Step 9: Motor Subassembly, cont’d

NEO Vortex Motors (8mm Keyed Shaft)

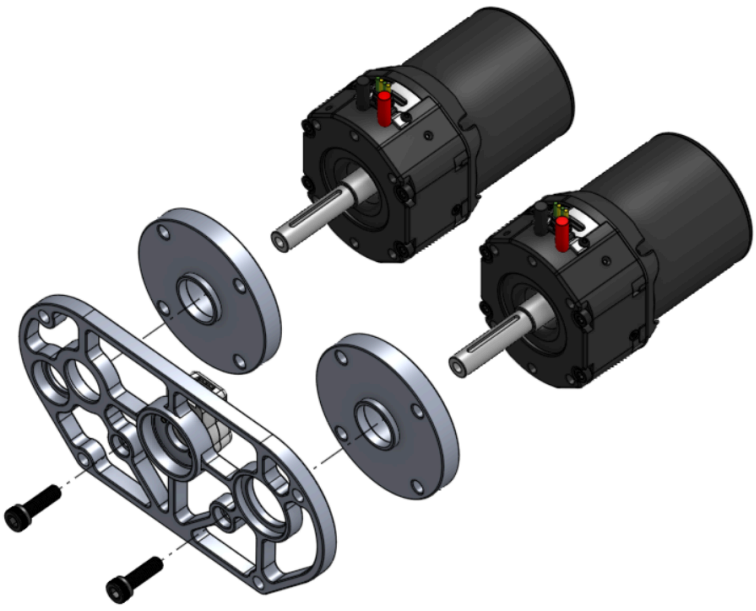
Secure the Motors and Motor Risers to the Main Plate with two #10-32 x 0.625” Socket Head Screws (included in the Motor Riser Kit). If no Motor Risers are used, then use the two #10-32 x 0.375” Socket Head Screws included in the swerve module kit.

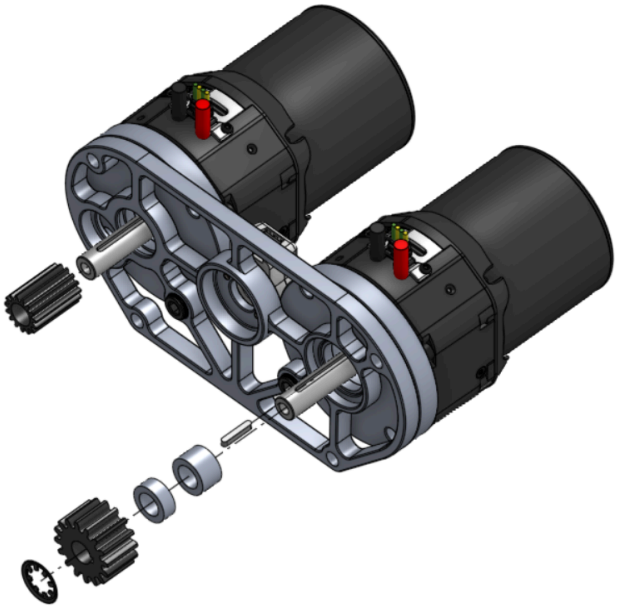
Drive Motor

Install the following onto the motor shaft in order: 0.313” Long Spacer, 0.188” Long Spacer, 2mm Machine Key, 16T Pinion Gear. Secure those parts with the 8mm ID Retaining Ring.

Steering Motor

Install the 15T gear (may be slip or press fit) 0.125” from the end of the shaft, using Loctite 609 Retaining Compound to fully secure the gear in place.



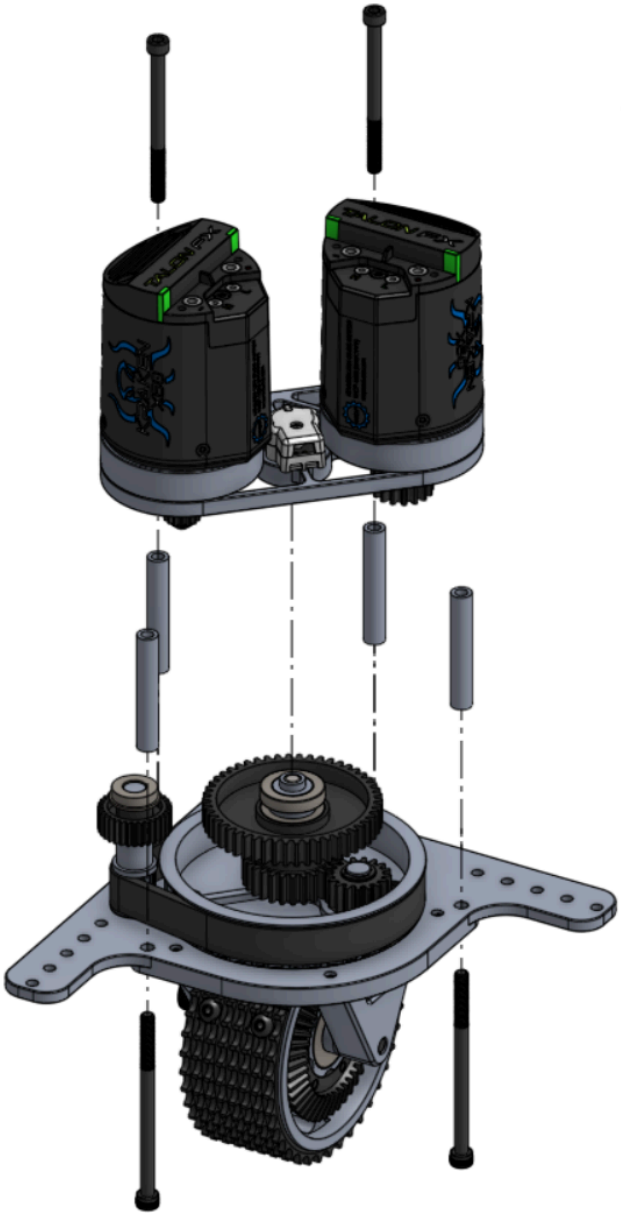


Step 10: Motor Plate Installation

Place the four 1.875” Long Spacers between the Main Plate and Motor Plate. Secure the Motor Plate by installing two #10-32 x 2.5” Socket Head Screws from the top, and two of the following screws (motor specific) from the bottom:

- If using the Motor Risers (screws are in the Motor Riser Kit)
  - Kraken X60: #10-32 x 3” Socket Head Screw
  - Falcon 500, NEO, or NEO Vortex: #10-32 x 2.75” Socket Head Screw
- If not using the Motor Risers (screws are in the swerve module kit)
  - #10-32 x 2.5” Socket Head Screw

The screws from the bottom pass through the Motor Plate and Motor Riser, and then screw into the motors.

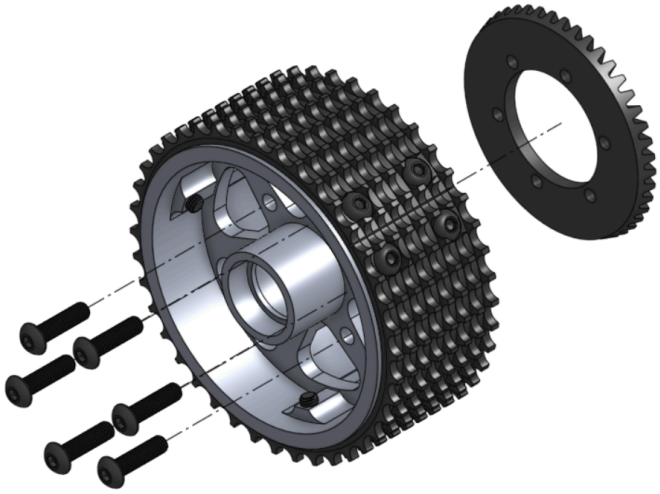


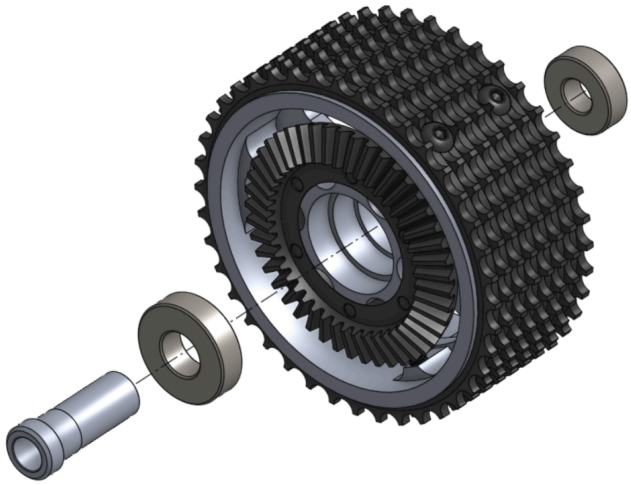
Step 11: Wheel Subassembly

The swerve module can be configured with a Billet Wheel or Colson Wheel.

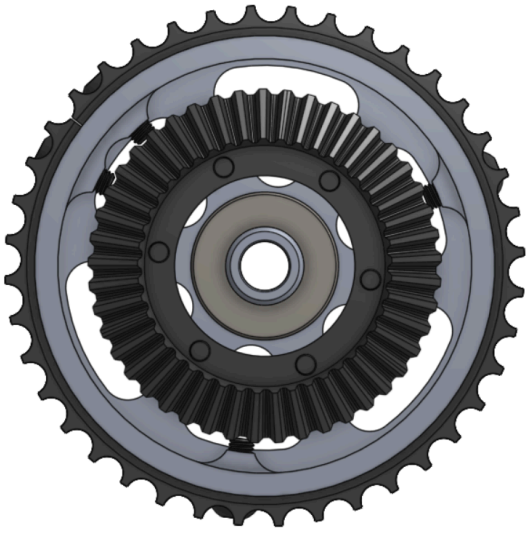
Billet Wheel

Secure the 45T Bevel Gear to the Wheel with six #10-32 x 0.75” Button Head Screws. Insert the R8ZZ Bearing (1.125” OD) and Wheel Spacer on the Bevel Gear side, and insert the R6ZZ Bearing (0.875” OD) on the opposite side.





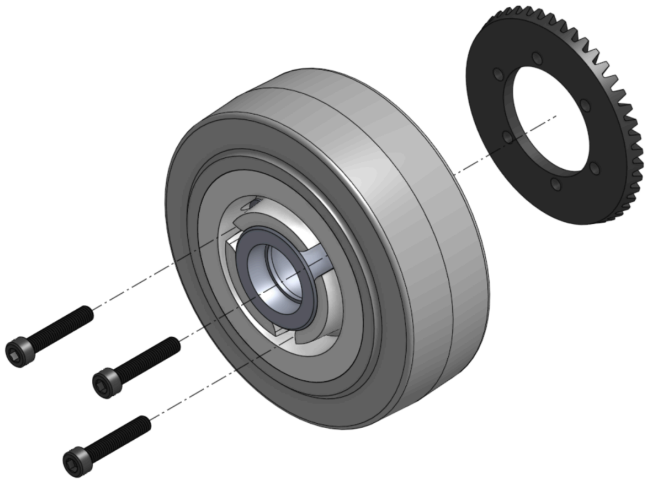


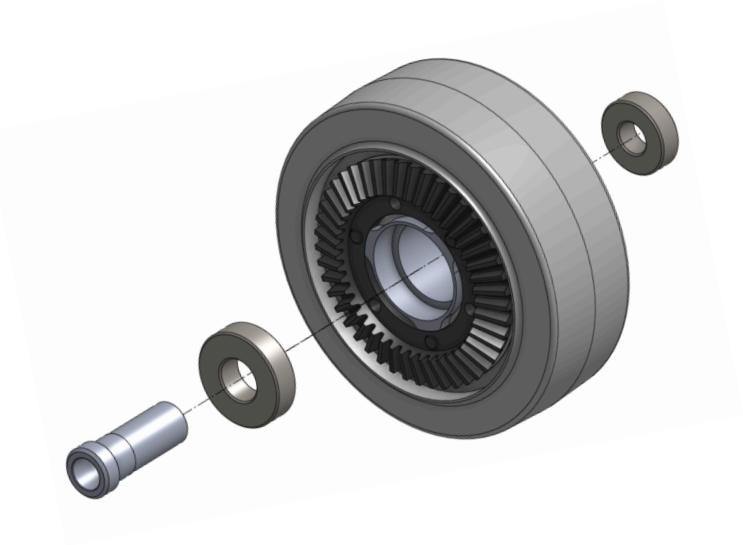


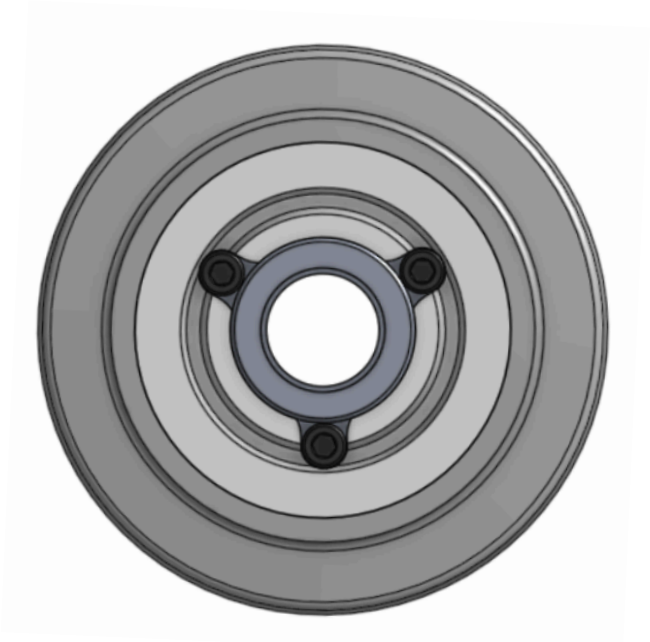
Step 11: Wheel Subassembly, cont’d

Colson Wheel

Secure the 45T Bevel Gear to the Wheel with three #10-32 x 1” Socket Head Screws. Insert the R8ZZ Bearing (1.125” OD) and Wheel Spacer on the Bevel Gear side, and insert the R6ZZ Bearing (0.875” OD) on the opposite side.



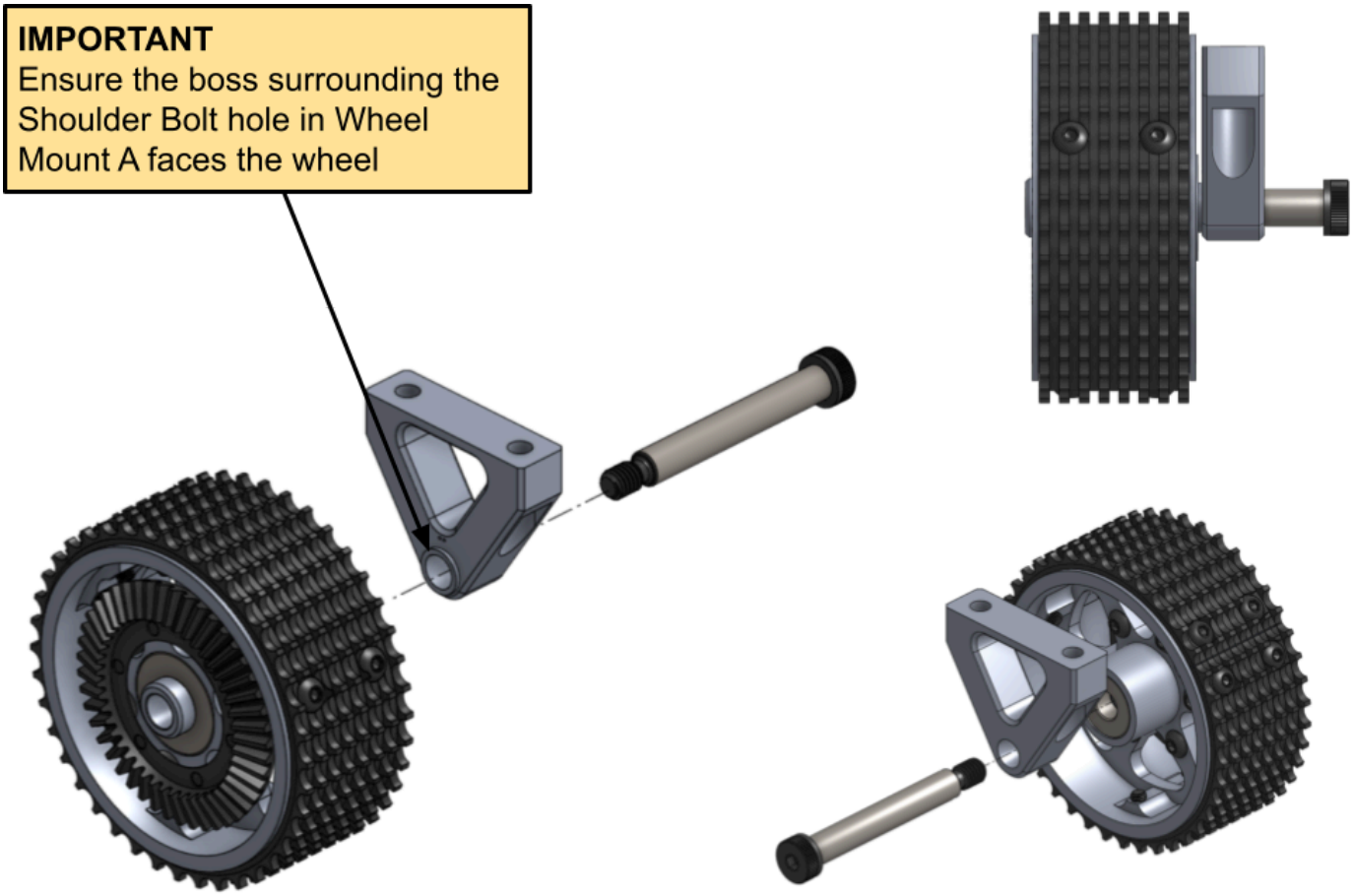






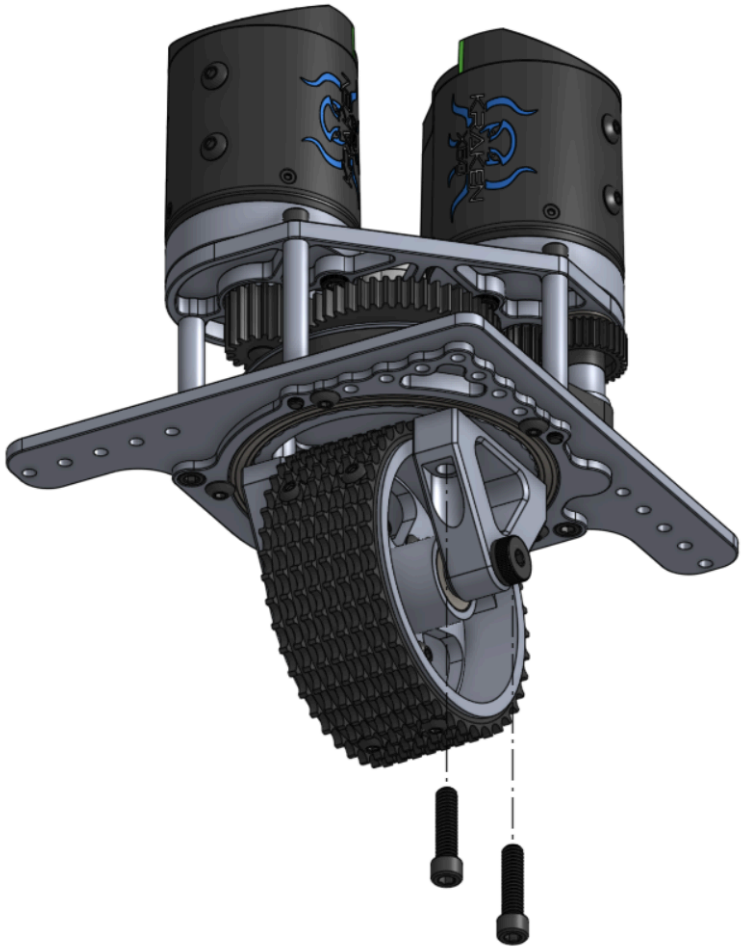
Step 12: Wheel Installation

Insert the Shoulder Bolt partially through Wheel Mount A and into the Wheel Spacer.



Step 12: Wheel Installation, cont’d

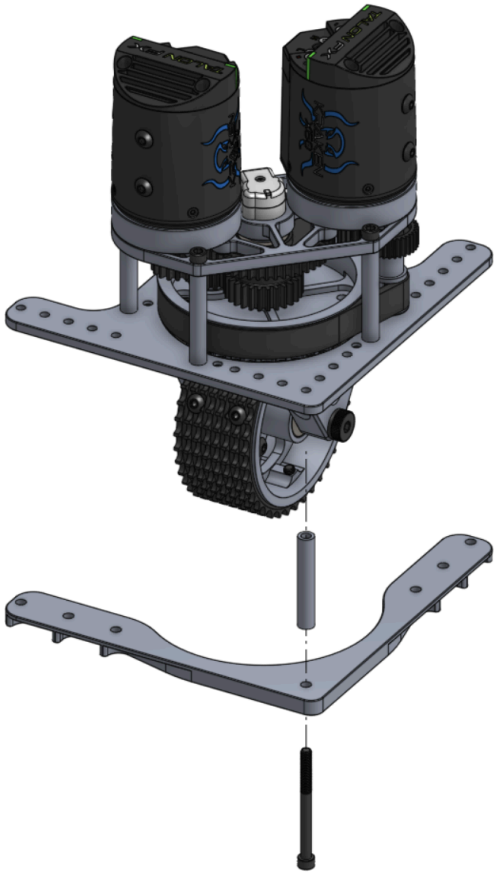
Attach the Wheel subassembly to the swerve module by securing Wheel Mount A to the Base Pulley with two 1/4”-20 x 1” Socket Head Screws, and the Shoulder Bolt into Wheel Mount B.

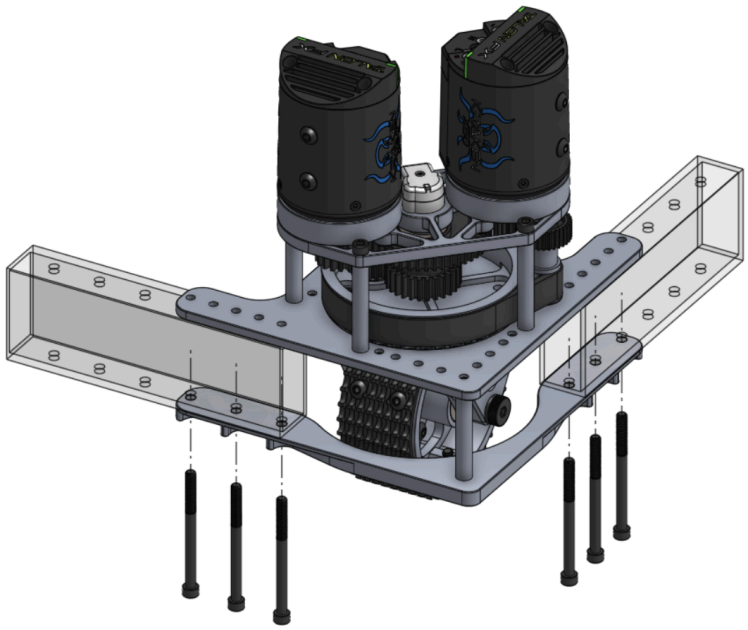


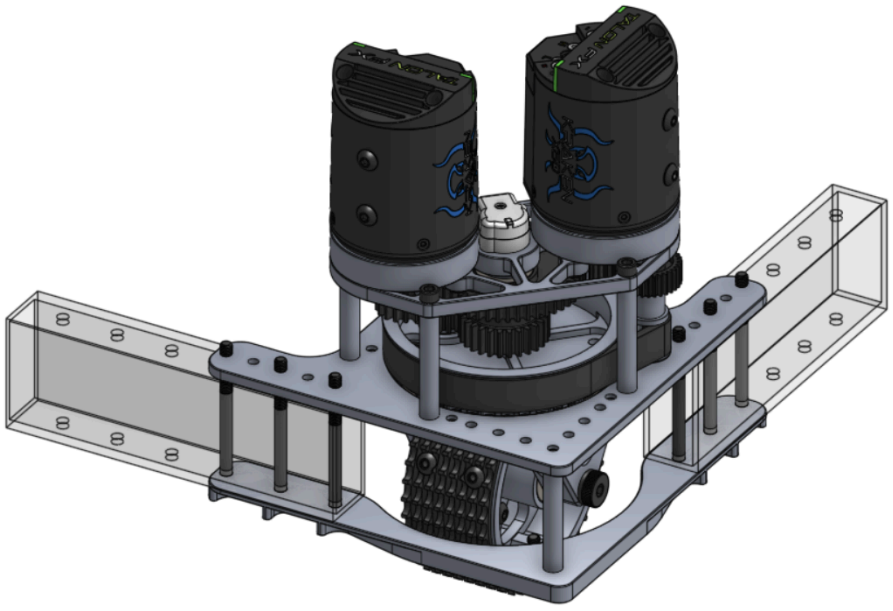
Step 13: Swerve Module Installation

Secure the #10 ID x 1.875” Long Spacer between the Main Plate and Bottom Plate with a #10-32 x 2.25” Socket Head Screw.

The swerve module is mounted to 2” x 1” rectangular tube with six #10-32 x 2.5” Socket Head Screws. Refer to the Appendix for additional rectangular tube details.







IMPORTANT Lubricate all gears with Lucas White Lithium, Lucas Red “N” Tacky, or equivalent grease BEFORE using the swerve module, and REGULARLY with use.

Appendix

Frame Tube Dimensions

The swerve module is designed to mount to a frame constructed from 2” x 1” rectangular tube. Each frame member (typically four total) needs to contain three vertical thru-holes at each end for attaching the swerve modules.

Pre-drilled tubing that only needs to be cut to length is available from multiple vendors, including AndyMark. Refer to the following drawing for the frame member size and mounting hole locations.

